AQA A-Level Further Mathematics (7367) — Topic 1.3

1.3: Further Algebra and Functions

1. Roots of Polynomials

Quadratic: $ax^2 + bx + c = 0$

If the roots are α and β:

$$\alpha + \beta = -\frac{b}{a} \qquad \alpha\beta = \frac{c}{a}$$

Cubic: $ax^3 + bx^2 + cx + d = 0$

If the roots are α, β, γ:

$$\alpha + \beta + \gamma = -\frac{b}{a} \qquad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} \qquad \alpha\beta\gamma = -\frac{d}{a}$$

Quartic: $ax^4 + bx^3 + cx^2 + dx + e = 0$

If the roots are α, β, γ, δ:

$$\alpha + \beta + \gamma + \delta = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a}$$

© RAG Learning · Not for redistribution

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -\frac{d}{a}$$

$$\alpha\beta\gamma\delta = \frac{e}{a}$$

AQA Exam Tip: Memorise the symmetric function relations for all three cases (quadratic, cubic, quartic). Use notation $\sum \alpha$, $\sum \alpha\beta$, $\sum \alpha\beta\gamma$ to save writing.

Symmetric Functions

Common symmetric functions that appear in AQA questions:

Expression	How to compute
$\alpha^2 + \beta^2$	$(\alpha+\beta)^2 - 2\alpha\beta$
$\alpha^3 + \beta^3$	$(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$
$\frac{1}{\alpha} + \frac{1}{\beta}$	$\dfrac{\alpha+\beta}{\alpha\beta}$
$\alpha^2\beta + \alpha\beta^2$	$\alpha\beta(\alpha+\beta)$
$\alpha^2 + \beta^2 + \gamma^2$	$(\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\beta\gamma+\gamma\alpha)$
$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$	$(\alpha\beta+\beta\gamma+\gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha+\beta+\gamma)$

Example: The equation $2x^3 - 5x^2 + x + 3 = 0$ has roots α, β, γ. Find $\alpha^2 + \beta^2 + \gamma^2$.

$$\alpha + \beta + \gamma = \frac{5}{2},\text{\quad } \alpha\beta + \beta\gamma + \gamma\alpha = \frac{1}{2}$$

$$\alpha^2 + \beta^2 + \gamma^2 = \left(\frac{5}{2}\right)^2 - 2\left(\frac{1}{2}\right) = \frac{25}{4} - 1 = \frac{21}{4}$$

2. Equations with Related Roots

Given an equation with roots α, β, γ, find an equation with roots that are functions of α, β, γ.

Method (substitution): Let w be the new root. Express x in terms of w, then substitute into the original equation.

Example: The equation $x^3 - 3x + 2 = 0$ has roots α, β, γ. Find an equation with roots $2\alpha, 2\beta, 2\gamma$.

Let $w = 2x$, so $x = w/2$. Substitute:
$\left(\frac{w}{2}\right)^3 - 3\left(\frac{w}{2}\right) + 2 = 0$
$\frac{w^3}{8} - \frac{3w}{2} + 2 = 0$
Multiply by 8: $w^3 - 12w + 16 = 0$

Example: The equation $x^3 - 4x + 1 = 0$ has roots α, β, γ. Find an equation with roots $\alpha + 1, \beta + 1, \gamma + 1$.

Let $w = x + 1$, so $x = w - 1$. Substitute:
$(w-1)^3 - 4(w-1) + 1 = 0$
$w^3 - 3w^2 + 3w - 1 - 4w + 4 + 1 = 0$
$w^3 - 3w^2 - w + 4 = 0$

AQA Exam Tip: Always substitute $x = f(w)$ where w is the new variable, not $x = g(w)$ (easy to get backwards). For roots $1/\alpha$, substitute $x = 1/w$, then multiply through to clear fractions.

3. Partial Fractions

Case 1: Linear Factors in Denominator

$$\frac{f(x)}{(ax+b)(cx+d)} = \frac{A}{ax+b} + \frac{B}{cx+d}$$

Case 2: Repeated Linear Factor

$$\frac{f(x)}{(ax+b)^2(cx+d)} = \frac{A}{ax+b} + \frac{B}{(ax+b)^2} + \frac{C}{cx+d}$$

Case 3: Irreducible Quadratic Factor

$$\frac{f(x)}{(ax^2+bx+c)(dx+e)} = \frac{Ax+B}{ax^2+bx+c} + \frac{C}{dx+e}$$

Used when the quadratic $ax^2 + bx + c$ cannot be factored over $\mathbb{R}$ (negative discriminant).

Improper Fractions

If the degree of the numerator is $\geq$ the degree of the denominator, divide first to get a polynomial plus a proper fraction:

Example: Express $\dfrac{x^3+2x^2-1}{x^2-1}$ in partial fractions.

Degree of numerator (3) $>$ degree of denominator (2): divide first.
$x^3 + 2x^2 - 1 = (x^2-1)(x+2) + (x+1)$

So $\dfrac{x^3+2x^2-1}{x^2-1} = x + 2 + \dfrac{x+1}{(x-1)(x+1)} = x + 2 + \dfrac{x+1}{(x-1)(x+1)}$
$= x + 2 + \dfrac{1}{x-1}$ (since $x \neq -1$)

Example: Decompose $\dfrac{5x+1}{(x+1)(2x-1)}$ into partial fractions.

$$\frac{5x+1}{(x+1)(2x-1)} = \frac{A}{x+1} + \frac{B}{2x-1}$$

Multiply through: $5x+1 = A(2x-1) + B(x+1)$

Set $x = -1$: $-4 = A(-3) \Rightarrow A = \dfrac{4}{3}$
Set $x = \frac{1}{2}$: $\dfrac{7}{2} = B\left(\dfrac{3}{2}\right) \Rightarrow B = \dfrac{7}{3}$

Result: $\dfrac{4/3}{x+1} + \dfrac{7/3}{2x-1} = \dfrac{4}{3(x+1)} + \dfrac{7}{3(2x-1)}$

AQA Exam Tip: AQA often uses partial fractions as a building block for integration or series questions. Always check your partial fractions by substituting back in or multiplying out.

4. Inequalities

Rational Inequalities

To solve $\dfrac{f(x)}{g(x)} > 0$ (or < 0, $\leq$, $\geq$): do NOT multiply both sides by $g(x)$ unless you know the sign of $g(x)$
.

Method:

1. Find critical values: zeros of numerator and denominator.
2. Draw a sign table/diagram — test the sign of $\dfrac{f(x)}{g(x)}$ in each interval.
3. State the solution set carefully, excluding values where denominator $= 0$.

Example: Solve $\dfrac{x-1}{x+2} < 2$.

$$\frac{x-1}{x+2} - 2 < 0 \Rightarrow \frac{x-1-2(x+2)}{x+2} < 0 \Rightarrow \frac{-x-5}{x+2} < 0 \Rightarrow \frac{x+5}{x+2} > 0$$

Critical values: $x = -5$ and $x = -2$.
Sign table:

Interval	$x+5$	$x+2$	Product
$x < -5$	$-$	$-$	$+$
$-5 < x < -2$	$+$	$-$	$-$
$x > -2$	$+$	$+$	$+$

Solution: $x < -5$ or $x > -2$.

Modulus Inequalities

Key results:

- $|f(x)| < a$ (where $a > 0$) means $-a < f(x) < a$
- $|f(x)| > a$ (where $a > 0$) means $f(x) > a$ or $f(x) < -a$
- $|f(x)| \leq |g(x)|$ means $f(x)^2 \leq g(x)^2$, i.e. $f^2 - g^2 \leq 0$

Example: Solve $|3x-1| < |x+3|$.

Square both sides: $(3x-1)^2 < (x+3)^2$
$9x^2 - 6x + 1 < x^2 + 6x + 9$
$8x^2 - 12x - 8 < 0 \Rightarrow 2x^2 - 3x - 2 < 0 \Rightarrow (2x+1)(x-2) < 0$

Critical values: $x = -\frac{1}{2}$ and $x = 2$.
Solution: $-\dfrac{1}{2} < x < 2$.

Example: Solve $|2x+1| \geq 5$.

$2x+1 \geq 5$ or $2x+1 \leq -5$
$x \geq 2$ or $x \leq -3$

5. Modulus Function Properties

Modulus function: $|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0 \end{cases}$

Properties:

- $|f(x)| \geq 0$ always
- $|ab| = |a||b|$
- $|a+b| \leq |a| + |b|$ (triangle inequality)
- $|a|^2 = a^2$

Graphs Involving Modulus

To sketch $y = |f(x)|$: sketch $y = f(x)$ first, then reflect any parts below the x-axis in the x-axis. To sketch $y = f(|x|)$: sketch for $x \geq 0$, then reflect in the y-axis.

AQA Exam Tip: When solving $|f(x)| = |g(x)|$, the two cases are $f(x) = g(x)$ and $f(x) = -g(x)$. Both must be solved and answers checked in the original equation (squaring can introduce extraneous solutions in some contexts).

6. Rational Functions and Oblique Asymptotes

A rational function $\dfrac{p(x)}{q(x)}$ where $\deg(p) = \deg(q) + 1$ has an **oblique (slant) asymptote**. Perform polynomial division to identify it.

Example: Find the asymptotes of $f(x) = \dfrac{x^2 + 3x + 1}{x + 1}$.

Divide: $x^2 + 3x + 1 = (x + 1)(x + 2) - 1$
So $f(x) = x + 2 - \dfrac{1}{x + 1}$.

As $x \to \pm\infty$, $f(x) \to x + 2$. Oblique asymptote: $y = x + 2$.
Vertical asymptote: $x = -1$.

7. Summary of Key Techniques

Topic	Key formula / method
Roots of quadratic	$\alpha + \beta = -b/a$, $\alpha\beta = c/a$
Roots of cubic	$\sum \alpha = -b/a$, $\sum \alpha\beta = c/a$, $\alpha\beta\gamma = -d/a$
Related roots	Substitute $x = f(w)$ into original equation
Partial fractions	Cover-up rule for distinct linear; compare coefficients for others
Rational inequalities	Rearrange to zero; sign table for critical values
Modulus inequalities	Square both sides or split into cases

8. Practice Checklist

- Can you compute all elementary symmetric functions for quadratic and cubic roots?
- Can you find an equation with roots that are simple functions of the original roots?
- Can you decompose proper and improper rational functions into partial fractions?
- Can you solve rational inequalities using sign tables?
- Can you solve modulus inequalities including $|f| < |g|$ type?

© RAG Learning · Not for redistribution

Newton's Identities and Power Sums

Let $p_k = \alpha^k + \beta^k + \gamma^k$ (power sums of roots). Newton's identities relate power sums to elementary symmetric polynomials $e_1 = \sum \alpha$, $e_2 = \sum \alpha\beta$, $e_3 = \alpha\beta\gamma$:

$$p_1 = e_1, \qquad p_2 = e_1 p_1 - 2e_2, \qquad p_3 = e_1 p_2 - e_2 p_1 + 3e_3$$

More generally, if the roots satisfy $\alpha^3 = a e_1 \alpha^2 + \dots$ (from the original equation), multiply by α^{n-3} and sum over roots to build a recurrence for p_n.

Example: The roots of $x^3 - 2x^2 + x - 3 = 0$ are α, β, γ. Find $p_2 = \alpha^2 + \beta^2 + \gamma^2$ and p_3. From Vieta: $e_1 = 2$, $e_2 = 1$, $e_3 = 3$. $p_2 = e_1^2 - 2e_2 = 4 - 2 = 2$. Each root satisfies $\alpha^3 = 2\alpha^2 - \alpha + 3$, so summing: $p_3 = 2p_2 - p_1 + 9 = 4 - 2 + 9 = 11$.

Solving Polynomial Inequalities Rigorously — Sign Table Method

To solve a rational inequality $f(x) > 0$:

1. Rearrange so the right-hand side is zero.
2. Find all **critical values**: zeros of numerator and zeros of denominator (excluded from the domain).
3. Mark critical values on a number line; test the sign of f in each interval.
4. Read off the solution, using strict or non-strict inequalities as appropriate.

Example: Solve $\dfrac{(x-1)(x+3)}{(x-2)^2} > 0$. Critical values: $x = -3$ (zero of numerator), $x = 1$ (zero of numerator), $x = 2$ (excluded).

Interval	$x - 1$	$x + 3$	$(x-2)^2$	Sign of f
$x < -3$	$-$	$-$	$+$	$+$
$-3 < x < 1$	$-$	$+$	$+$	$-$
$1 < x < 2$	$+$	$+$	$+$	$+$
$x > 2$	$+$	$+$	$+$	$+$

© RAG Learning · Not for redistribution

Solution: $x < -3$ or $x > 1$ (excluding $x = 2$).

Important: Never multiply through by an expression that could be negative. The sign table avoids this entirely and is the recommended AQA method.

Improper Algebraic Fractions — Partial Fraction Examples

A fraction is **improper** when deg(numerator) ≥ deg(denominator). Always perform polynomial long division first, then decompose the proper remainder.

Example: Express $\dfrac{2x^3 + 3x^2 - 2}{x^2 + x - 2}$ in partial fractions. *Step 1 — Long division:* Dividing gives quotient $2x + 1$, remainder $3x$. (Verify: $(2x+1)(x^2+x-2) + 3x = 2x^3 + 3x^2 - 2$ ✓) *Step 2 — Partial fractions of* $\dfrac{3x}{(x-1)(x+2)}$:

$$\frac{3x}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

$x = 1$: $A = 1$. $x = -2$: $B = 2$. *Answer:* $2x + 1 + \dfrac{1}{x-1} + \dfrac{2}{x+2}$.

Repeated factor: Express $\dfrac{5x+3}{(x+1)^2(x-2)}$ in partial fractions.

$$= \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-2}$$

$x = -1$: $-2 = -3B \Rightarrow B = \frac{2}{3}$. $x = 2$: $13 = 9C \Rightarrow C = \frac{13}{9}$. x^2 coefficient: $A + C = 0 \Rightarrow A = -\frac{13}{9}$. Check x^0: $-2A - 2B + C = \frac{26}{9} - \frac{4}{3} + \frac{13}{9} = 3$ ✓.

AQA Exam Technique — Further Algebra

Key tips:

- **Vieta's formulas:** Memorise the alternating signs for cubic: sum $= -b/a$, pairs $= +c/a$, product $= -d/a$ for $ax^3 + bx^2 + cx + d = 0$.
- **Related equations:** To find the equation with roots $f(\alpha)$, let $y = f(\alpha)$, rearrange to $\alpha = g(y)$, substitute into the original equation, then multiply out.
- **Partial fractions:** Always verify by recombining over a common denominator. Cover-up rule only works for distinct linear factors.
- **Modulus $|f| < |g|$:** Square both sides to get $f^2 - g^2 < 0$, then factorise as $(f - g)(f + g) < 0$ and use a sign table.
- **Power sums:** Use Newton's identity $p_2 = (\sum \alpha)^2 - 2\sum \alpha\beta$; always write this down before starting.

Type	Method	Watch out for
Power sums p_k	Newton's identities or recurrence	Sign errors in Vieta
$\lvert f(x)\rvert < k$	Solve $-k < f < k$	Forgetting the negative case
$\lvert f\rvert < \lvert g\rvert$	Square: $(f - g)(f + g) < 0$	Squaring valid when both sides ≥ 0
Rational $f(x) > 0$	Sign table at critical values	Not excluding denominator zeros
Improper PF	Divide first, then partial fractions	Skipping the division step

Finding Equations with Transformed Roots — Worked Examples

Example 1: The roots of $x^3 - 3x + 1 = 0$ are α, β, γ. Find the equation with roots $\alpha + 2, \beta + 2, \gamma + 2$. Let $y = x + 2$, so $x = y - 2$. Substitute:

$$(y - 2)^3 - 3(y - 2) + 1 = 0$$

$$y^3 - 6y^2 + 12y - 8 - 3y + 6 + 1 = 0$$

$$y^3 - 6y^2 + 9y - 1 = 0$$

Example 2: The roots of $2x^3 - x^2 + 3x - 4 = 0$ are α, β, γ. Find the equation with roots $1/\alpha, 1/\beta, 1/\gamma$. Let $y = 1/x$, so $x = 1/y$. Substitute and multiply through by y^3:

$$2(1/y)^3 - (1/y)^2 + 3(1/y) - 4 = 0$$

$$\text{Multiply by } y^3: \quad 2 - y + 3y^2 - 4y^3 = 0$$

$$4y^3 - 3y^2 + y - 2 = 0$$

Sketching Rational Functions

Key features to find before sketching $y = \dfrac{f(x)}{g(x)}$:

1. **Vertical asymptotes:** where $g(x) = 0$ (and $f(x) \neq 0$)
2. **Horizontal/oblique asymptotes:** behaviour as $x \to \pm\infty$ (long division if degrees equal or numerator higher)
3. **Intersections with axes:** y-intercept at $x = 0$; x-intercepts where $f(x) = 0$
4. **Stationary points:** differentiate and set $y' = 0$

Example: Sketch $y = \dfrac{x^2 - 1}{x - 2}$. Long division: $y = x + 2 + \dfrac{3}{x - 2}$ (oblique asymptote $y = x + 2$). Vertical asymptote: $x = 2$. Roots: $x = \pm 1$. y-intercept: $y = \frac{-1}{-2} = \frac{1}{2}$. For large $|x|$, $y \approx x + 2$.

Proof by Contradiction — Further Examples

Prove there are infinitely many primes. Suppose there are finitely many primes: $p_1, p_2, \ldots, p_n$. Let $N = p_1 p_2 \cdots p_n + 1$. Then $N > 1$ so has a prime factor p. But p divides N and divides $p_1 p_2 \cdots p_n$, so p divides $N - p_1 p_2 \cdots p_n = 1$. No prime divides 1 — contradiction. So there are infinitely many primes. ✓

Symmetry Properties and Odd/Even Functions

Property	Condition	Example
Even function	$f(-x) = f(x)$	$\cos x$, x^2, $\cosh x$
Odd function	$f(-x) = -f(x)$	$\sin x$, x^3, $\sinh x$
Periodic with period T	$f(x+T) = f(x)$	$\sin x$ (period 2π)

For partial fractions of odd/even integrands:

- Even function: all terms in PF are even — only even powers of x in numerators
- Odd function: all terms are odd — can save computation by checking symmetry

Vieta's Formulas — Three Roots

For cubic $ax^3 + bx^2 + cx + d = 0$ with roots α, β, γ:

$$\alpha + \beta + \gamma = -\frac{b}{a}, \qquad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \qquad \alpha\beta\gamma = -\frac{d}{a}$$

Example — Sum of squares: Find $\alpha^2 + \beta^2 + \gamma^2$ for roots of $x^3 - 3x^2 + 5x - 2 = 0$.
$\alpha + \beta + \gamma = 3,\ \alpha\beta + \beta\gamma + \gamma\alpha = 5,\ \alpha\beta\gamma = 2.$
$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = 9 - 10 = -1.$

Example — Sum of cubes: Using Newton's identity $p_3 = e_1 p_2 - e_2 p_1 + 3e_3$ where $p_k = \alpha^k + \beta^k + \gamma^k$, e_1, e_2, e_3 are elementary symmetric polynomials.
$p_1 = 3,\ p_2 = -1$ (above). $p_3 = 3 \cdot (-1) - 5 \cdot 3 + 3 \cdot 2 = -3 - 15 + 6 = -12.$

Partial Fractions — Complete Methods

All cases:

Denominator form	Partial fraction form
$(ax+b)(cx+d)$	$\dfrac{A}{ax+b}+\dfrac{B}{cx+d}$
$(ax+b)^2$	$\dfrac{A}{ax+b}+\dfrac{B}{(ax+b)^2}$
$(ax+b)(cx^2+d)$ irreducible	$\dfrac{A}{ax+b}+\dfrac{Bx+C}{cx^2+d}$
$(ax+b)^3$	$\dfrac{A}{ax+b}+\dfrac{B}{(ax+b)^2}+\dfrac{C}{(ax+b)^3}$

Example — Irreducible quadratic factor: $\dfrac{2x^2+x+1}{(x-1)(x^2+x+1)}$.

Write $=\dfrac{A}{x-1}+\dfrac{Bx+C}{x^2+x+1}$.

At $x=1$: $\dfrac{4}{3}=\dfrac{A\cdot 3}{3}\Rightarrow A=\dfrac{4}{3}$.

Equating x^2: $2=A+B\Rightarrow B=\dfrac{2}{3}$. Equating constants: $1=A-C\Rightarrow C=\dfrac{1}{3}$.

Result: $\dfrac{4/3}{x-1}+\dfrac{(2x+1)/3}{x^2+x+1}$.

Polynomial Long Division

For an improper fraction $\dfrac{f(x)}{g(x)}$ where $\deg f\geq\deg g$: divide to get $q(x)+\dfrac{r(x)}{g(x)}$ where $\deg r<\deg g$.

Example: Decompose $\dfrac{x^3+2x-1}{x^2-1}$ into partial fractions.

Long division: $x^3+2x-1=x(x^2-1)+3x-1$. So $\dfrac{x^3+2x-1}{x^2-1}=x+\dfrac{3x-1}{x^2-1}$.

$\dfrac{3x-1}{(x-1)(x+1)} = \dfrac{A}{x-1} + \dfrac{B}{x+1}$. $A = 1, B = 2$.
Final: $x + \dfrac{1}{x-1} + \dfrac{2}{x+1}$.

Inequalities Involving Modulus

- $|f(x)| < a \Leftrightarrow -a < f(x) < a$
- $|f(x)| > a \Leftrightarrow f(x) > a$ or $f(x) < -a$
- $|f(x)| < |g(x)| \Leftrightarrow f(x)^2 < g(x)^2$ (useful: square both sides)

Example 1: Solve $|2x-3| < 5$.
$-5 < 2x - 3 < 5 \Rightarrow -2 < 2x < 8 \Rightarrow -1 < x < 4$.

Example 2: Solve $|x-1| > |x+2|$.
Square: $(x-1)^2 > (x+2)^2 \Rightarrow x^2 - 2x + 1 > x^2 + 4x + 4 \Rightarrow -6x > 3 \Rightarrow x < -\frac{1}{2}$.

The Fundamental Theorem of Algebra

Every polynomial of degree n with complex coefficients has exactly n roots (counted with multiplicity) in $\mathbb{C}$.
Consequence for real polynomials: Complex roots occur in conjugate pairs. A degree n real polynomial has either 0, 2, 4, ... complex roots, and the remaining roots are real.

Example: A cubic $f(x)$ with real coefficients has root $2 + 3i$. Find all roots, given that the product of roots is 26.
Since coefficients are real, $2 - 3i$ is also a root. Let the third root be r.
Product: $(2+3i)(2-3i) \cdot r = 13r = 26 \Rightarrow r = 2$.
Roots: $2,\ 2+3i,\ 2-3i$.

Rational Inequalities — Worked Examples

Example 1: Solve $\dfrac{x+1}{x-2} > 3$.

$\dfrac{x+1}{x-2} - 3 > 0 \Rightarrow \dfrac{x+1-3(x-2)}{x-2} > 0 \Rightarrow \dfrac{-2x+7}{x-2} > 0$.

Critical values: $x = 2, x = 7/2$. Sign table:

$x < 2$: $-/- = +$. \(27/2\): $-/+ = -$... wait: $-2x + 7 > 0$ when $x < 7/2$, and $x - 2 > 0$ when $x > 2$.

Solution: $2 < x < 7/2$.

Example 2: Solve $\dfrac{1}{x} \geq \dfrac{1}{x+1}$ (without multiplying by x since sign unknown).

$\dfrac{1}{x} - \dfrac{1}{x+1} \geq 0 \Rightarrow \dfrac{x+1-x}{x(x+1)} \geq 0 \Rightarrow \dfrac{1}{x(x+1)} \geq 0$.

This requires $x(x+1) > 0$ (since numerator is 1 > 0).

$x(x+1) > 0$ when $x < -1$ or $x > 0$. Also $x \neq 0, x \neq -1$.

Solution: $x < -1$ or $x > 0$.

Roots of Polynomial Equations — Further Transformations

If α, β, γ are roots of $f(x) = 0$, then:

- Roots $k\alpha, k\beta, k\gamma$: substitute $x \to x/k$ in f, i.e. write $f(x/k) = 0$.
- Roots $\alpha + k, \beta + k, \gamma + k$: substitute $x \to x - k$.
- Roots $1/\alpha, 1/\beta, 1/\gamma$: substitute $x \to 1/x$ and multiply through by x^n.
- Roots $\alpha^2, \beta^2, \gamma^2$: substitute $x \to \sqrt{x}$ (or $x \to -\sqrt{x}$) and eliminate roots.

Example — Roots α^2: $x^2 - 3x + 5 = 0$ has roots α, β. Find equation with roots α^2, β^2.

$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 9 - 10 = -1$. $\alpha^2\beta^2 = (\alpha\beta)^2 = 25$.

Equation: $x^2 - (-1)x + 25 = 0 \Rightarrow x^2 + x + 25 = 0$.

Example — Roots $2\alpha + 1$: $2x^3 - 3x^2 + x - 4 = 0$ has roots α, β, γ. Find equation with roots $2\alpha + 1$.

Substitute $y = 2x + 1 \Rightarrow x = \dfrac{y-1}{2}$:

$2\left(\dfrac{y-1}{2}\right)^3 - 3\left(\dfrac{y-1}{2}\right)^2 + \dfrac{y-1}{2} - 4 = 0.$

Multiply through by 8: $(y-1)^3 - 6(y-1)^2 + 4(y-1) - 32 = 0.$

Expand: $y^3 - 3y^2 + 3y - 1 - 6y^2 + 12y - 6 + 4y - 4 - 32 = 0.$

$y^3 - 9y^2 + 19y - 43 = 0.$

Graph Sketching — Rational Functions

To sketch $y = \dfrac{f(x)}{g(x)}$:

1. Find vertical asymptotes: $g(x) = 0$.
2. Find horizontal/oblique asymptote: compare degrees of numerator and denominator.
3. Find axis intercepts: $y(0)$ and $f(x) = 0$.
4. Find stationary points: differentiate using quotient rule.
5. Determine behaviour near asymptotes (sign of y in each region).

Example — Sketch $y = \dfrac{x}{x^2 - 1}$:

Vertical asymptotes: $x = \pm 1$. Horizontal: $y \to 0$ as $x \to \pm\infty$ (degree of denom > degree of numer).

Intercept: $(0, 0)$. Odd function (rotational symmetry about origin).

$y' = \dfrac{(x^2-1) - x \cdot 2x}{(x^2-1)^2} = \dfrac{-x^2 - 1}{(x^2-1)^2} < 0$ always (where defined). So no stationary points; function is always decreasing on each branch.

Algebraic Inequalities — Key Principles

- Never multiply an inequality by an expression that could be negative (unless you consider cases or use the squared method).
- **Safe method:** Always bring everything to one side and factorise/find sign by sign table.
- For $f(x)/g(x) > 0$: find roots of f and g, make sign table, find where quotient is positive.
- $(x-a)^2 \geq 0$ always: use this to prove inequalities like $x^2 + 1 > 2x - 1$.
- AM-GM: $\dfrac{a+b}{2} \geq \sqrt{ab}$ for $a, b \geq 0$. Useful in proof questions.

AQA Further Algebra — Exam Technique

- **Partial fractions:** Show all working for finding A, B, C. Use substitution for linear factors; equate coefficients for others.
- **Roots questions:** Build up expressions from e_1, e_2, e_3 step by step. Don't try to find the actual roots.
- **Inequalities:** Never multiply by a variable. Bring everything to one side. Present as a sign table or clear argument.
- **Sketches:** AQA expects: asymptotes labelled; intercepts shown; correct behaviour at extremes. A small table of values near asymptotes earns marks.
- **Proving polynomial identities:** Use the fact that if two polynomials of degree n agree at $n+1$ points, they are identical.

Quick Summary: Symmetric Functions of Roots

For $ax^3 + bx^2 + cx + d = 0$ with roots α, β, γ:

Expression	Formula
$\sum \alpha$	$-b/a$

$\sum \alpha\beta$	c/a
$\alpha\beta\gamma$	$-d/a$
$\sum \alpha^2$	$(\sum \alpha)^2 - 2\sum \alpha\beta$
$\sum \alpha^2\beta$	$(\sum \alpha)(\sum \alpha\beta) - 3\alpha\beta\gamma$
$\sum \alpha^3$	$(\sum \alpha)^3 - 3(\sum \alpha)(\sum \alpha\beta) + 3\alpha\beta\gamma$

© RAG Learning · Not for redistribution